

Housing Rental Policies & Rent Growth Trends

Chicago Metropolitan Area

Yice Zhang, Ke Gao · Data Visualization · Final Project

Motivation

- Rents rose +21.9% from 2015 to 2024
- Post-pandemic surge: +12.1% in just 4 years (2019–2023)
- 83% of Cook County census tracts have majority cost-burdened renters
- Cost burden = spending >30% of income on rent (HUD threshold)

"A majority of Chicago renters spend more than 30% of their income on housing, but the crisis is not spread evenly across the city."

Research Questions

01

Temporal Dynamics

How have rents evolved across Chicago ZIP codes from 2015 to 2026, and how did COVID-19 policy shape that trajectory?

02

Spatial Inequality

Which census tracts bear the highest rent burdens, and is geographic clustering visible?

03

Policy Relevance

Do rent trends and burden concentrations align with key Chicago housing policy events?

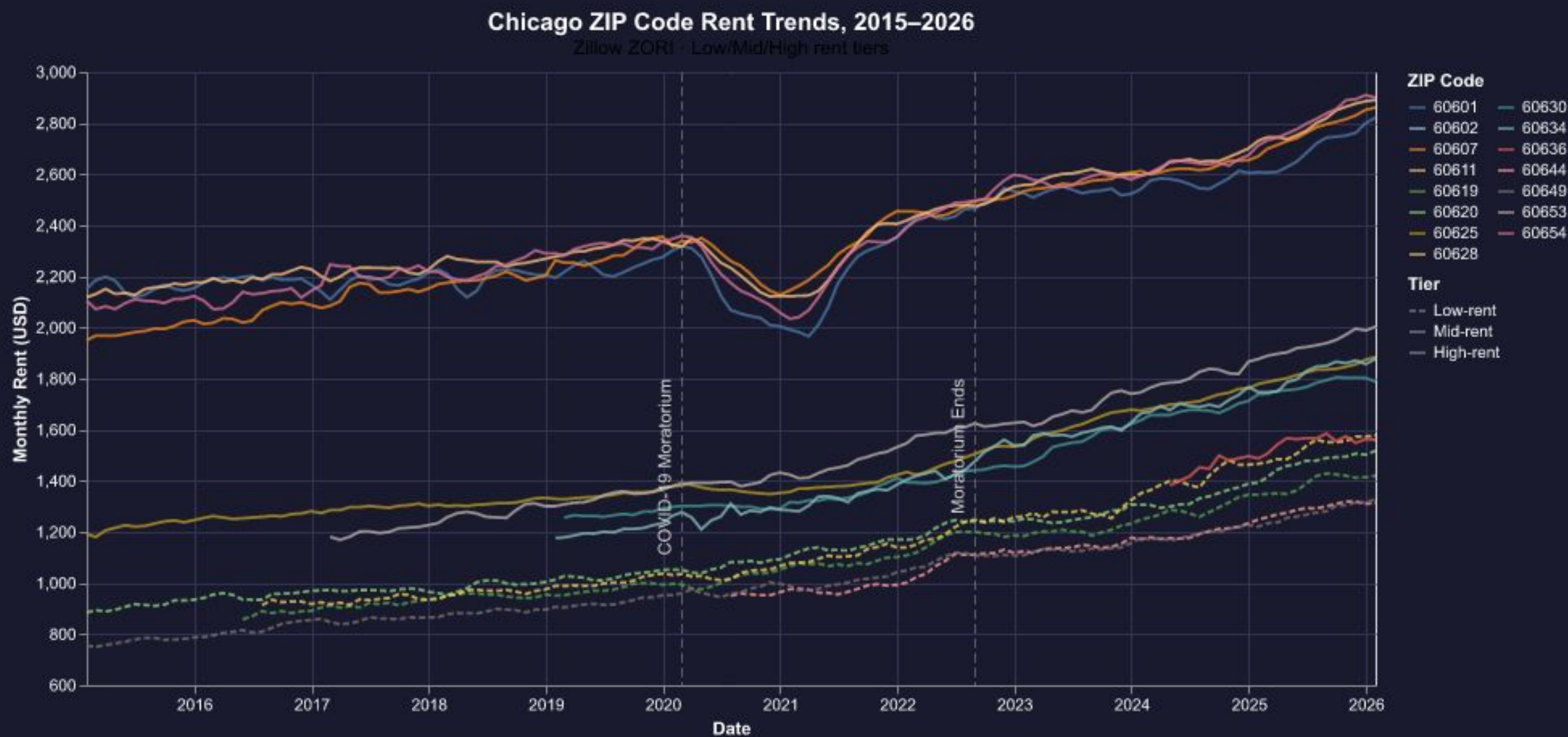
Data Sources

Two independent datasets merged by geography

Dataset	Source	Unit	Coverage
ACS B25064 – Median Gross Rent	U.S. Census Bureau	Census tract	1,254 Cook County tracts
ACS B19013 – Median Household Income	U.S. Census Bureau	Census tract	1,254 Cook County tracts
ACS B25070 – Rent Burden Bins	U.S. Census Bureau	Census tract	1,254 Cook County tracts
Zillow ZORI – Monthly Rent Index	Zillow Research	ZIP code	206 Chicago metro ZIPs · 2015–2026

Finding 1: Rent Trends Over Time (2015–2026)

Zillow ZORI · 15 Chicago ZIP codes across Low / Mid / High rent tiers



Finding 1: Key Takeaways

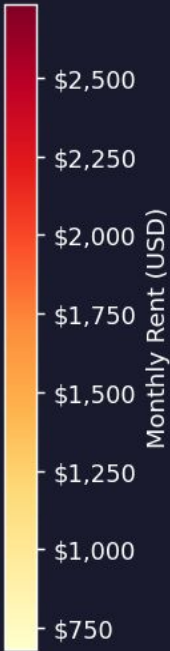
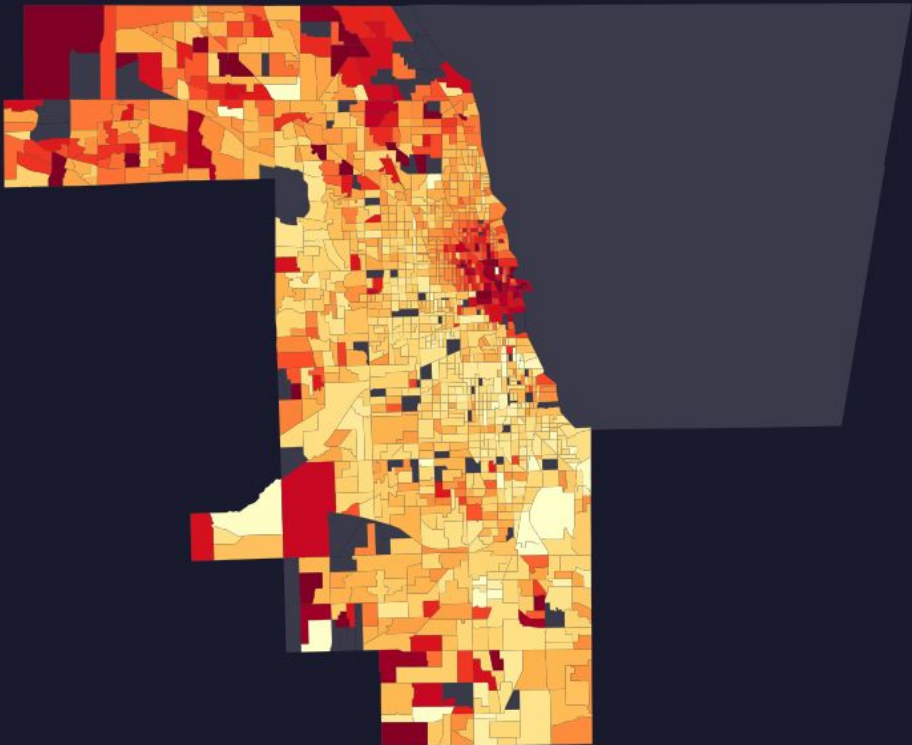
Before COVID (2015–2019)	COVID Period (2020–2021)	Post-Moratorium Surge (2022–2024)
• Modest growth: +3.2% over 4 years • Market broadly stable across tiers	• Rents dipped: \$1,532 → \$1,521 • Eviction moratorium suppressed re-pricing	• 2023 average: \$1,728 (+12.1% vs 2019) • 2024 average: \$1,820 (+21.9% vs 2015) • High-rent ZIPs recovered faster • Intra-city rent gap widened

Finding 2: Geographic Inequality

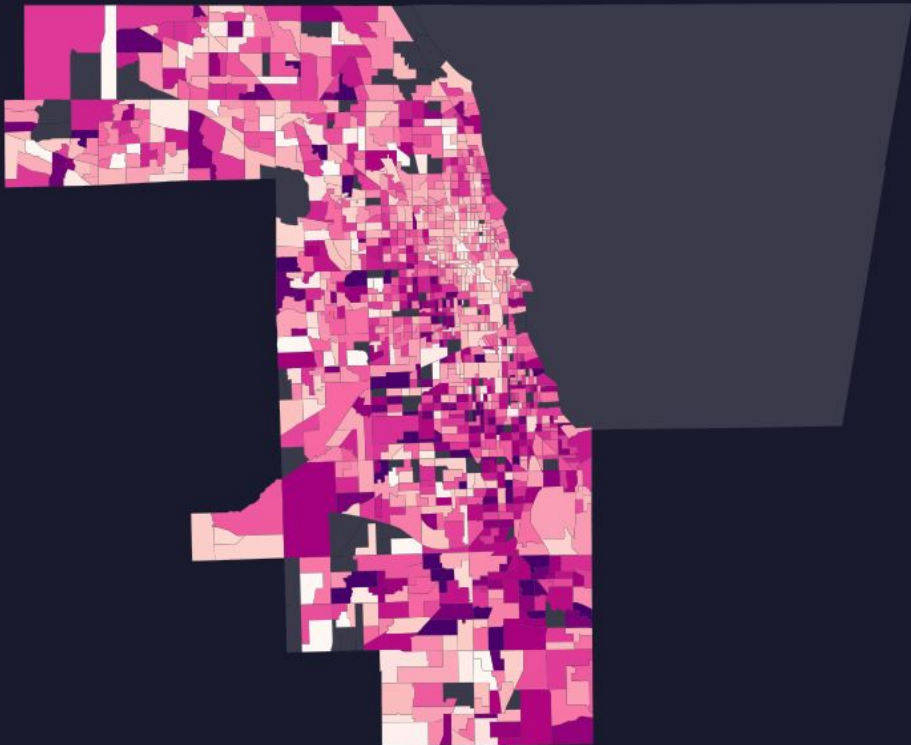
Cook County census tracts · 2024 ACS · Left: median rent | Right: % cost-burdened

Cook County Housing Affordability · Census Tract Level

Median Gross Rent (2024 ACS)



Share of Cost-Burdened Renters (2024 ACS)



Finding 2: Key Takeaways

Rent Level (left map)

- High rents cluster in Near North Side, Lincoln Park, Lakeview (>\$2,000/mo)
- South & West Side tracts often below \$900/mo
- Clear North–South gradient

Cost Burden (right map)

- Severe burden $\neq$ highest-rent areas
- Concentrated in South/West Side (Englewood, Austin, Garfield Park)
- 40.7% of tracts: majority of renters are cost-burdened
- Driven by income inadequacy, not just high rents

83.4%

of tracts exceed
HUD 30% threshold

40.7%

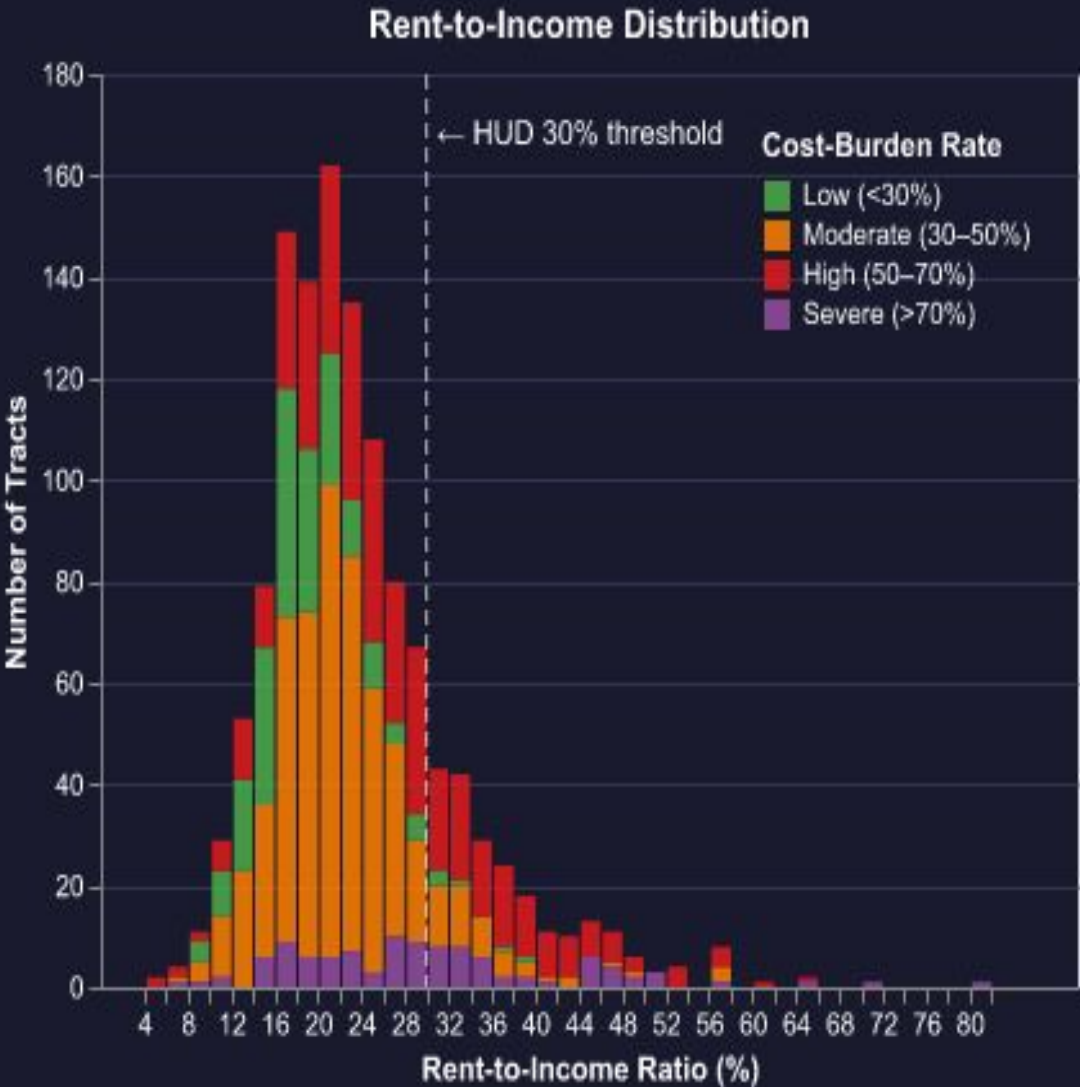
of tracts have >50%
renters burdened

\$1,100+

median rent gap
North vs South Side

Finding 3: Affordability Distribution

Cook County census tracts · 2024 ACS · Left: RTI histogram | Right: cumulative burden rate



Finding 3: Key Takeaways

- Median rent-to-income ratio across Cook County: 21.9% — looks OK at the aggregate...
- But 83.4% of tracts have >30% of their renters cost-burdened
- 40.7% of tracts exceed the 50% threshold — the majority of renters in those tracts
- "Severe" tracts (>70% burdened) appear across the full RTI range: burden is driven by low income, not just high rent
- Only ~17% of tracts fall below the HUD affordability threshold

Interactive Dashboard

Live Streamlit App — link: <https://datavisualizationfinalproject-m7rnhvqfe9uemeai2vqwjt.streamlit.app/>

Tab 1 Rent Trends

- Multi-select ZIP codes & cities
- Year range slider
- Policy event overlay toggle
- Summary stats table

Tab 2 Choropleth Map

- Switch: rent / RTI / burden rate
- Adjustable color scheme
- Top 10 burdened tracts table

Tab 3 Distribution

- RTI histogram (interactive)
- Cumulative burden ECDF
- Rent vs. burden scatter plot



Remember to wake up the app the night before the presentation.

Policy Implications

Finding	Policy Recommendation
+12% rent surge after moratorium ends	Stronger tenant relocation assistance & just-cause eviction protections
Severe burden in South/West corridors	Target vouchers & affordable unit production to high-burden areas
83% of tracts exceed HUD threshold	Expand rental assistance eligibility — current scale is insufficient
North Side rent appreciation gap	Anti-displacement zoning overlays in rapidly appreciating neighborhoods

Chicago's Affordable Requirements Ordinance (ARO) needs stronger income targeting and enforcement.

Conclusion

1

Rents rose 21.9% from 2015–2024, with the sharpest acceleration post-moratorium (2022–2024).

2

Geographic inequality is severe: high rents in the North, but the worst burden is concentrated in South/West Side neighborhoods due to low incomes.

3

Cost burden is the norm: 83% of Cook County tracts fail the HUD affordability standard — a systemic failure, not an isolated problem.

4

Both supply and income policies are needed: market-rate construction alone won't reach the communities that need help most.

Thank You

Data: ACS 2024 · Zillow ZORI 2015–2026 · Census TIGER/Line 2024

Streamlit Dashboard: <https://datavisualizationfinalproject-m7rnhvqfe9uemeai2vqwjt.streamlit.app/>